

Introduction to Robotics

Lecture 14: Lagrangian dynamics

- It is often **implicitly** assumed that the robots' links have negligible mass, at least compared to the actuation power of their actuators. In this case, the **kinematic approach** to motion describes the actual physical system relatively well.
- We look in this lecture into the effects of **non-negligible masses**, and thus **inertia**, on the dynamics of robots.
- **Inverse dynamics**: determine the torques yielding a given trajectory $(\theta, \dot{\theta}, \ddot{\theta})$:

$$\tau = M(\theta)\ddot{\theta} + h(\theta, \dot{\theta})$$

- **Forward dynamics**: determines the trajectory $\theta(t)$ given τ :

$$\ddot{\theta} = M^{-1}(\theta)(\tau - h(\theta, \dot{\theta})).$$

The trajectory is a solution of the above ODE with appropriate initial state.

- $M(\theta)$ is called the **mass matrix**

- We first review the **Lagrangian approach** to determine the dynamics of a rigid body.
- Denote by $q \in \mathbb{R}^n$ the so-called **generalized coordinates** of the system. These are a set of coordinates describing its state.
- From the generalized coordinates, we define the **generalized forces** $f \in \mathbb{R}^n$. These are the forces on the system that “act” on the generalized coordinates.
- The pair needs to be **consistently chosen** so that the *power* dissipated by the system is $f^\top \dot{q}$.
- Mathematically, generalized forces are the dual to the (derivatives) of the generalized coordinates. We do not go into these details in this course.
- Denote by $K(q, \dot{q})$ the **kinetic** energy of the system, and by $P(q)$ its **potential energy**. (Recall that potential energy does *not* depend on $\dot{q}$). The **Lagrangian** of the system is defined as

$$L(q, \dot{q}) = K(q, \dot{q}) - P(q)$$

- From the Lagrangian of the system, we obtain the equations of motion through the principle of **least action**, which yields the **Euler-Lagrange** equations:

$$f = \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q}.$$

- Consider a particle of mass m constrained to move on the vertical line.
- A generalized coordinate is its height $x \in \mathbb{R}$.
- Suppose that an external force f is applied on it, and force due to gravity is given by mg .
- The Lagrangian is then

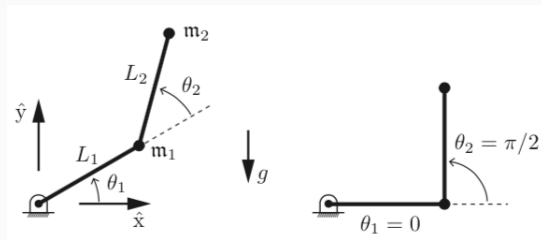
$$L(x, \dot{x}) = K(x, \dot{x}) - P(x) = \underbrace{\frac{1}{2}m\dot{x}^2}_{\text{kin. en.}} - \underbrace{mgx}_{\text{pot. en.}}.$$

- The equations of motion are given by

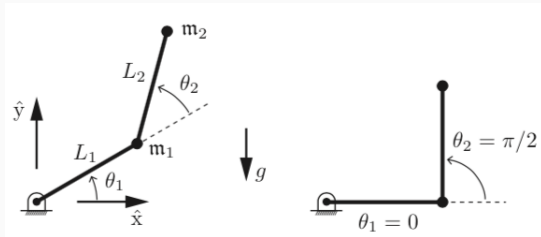
$$f = \frac{d}{dt}(m\dot{x}) - (-mg) = m\ddot{x} + mg.$$

- We obtain the same equations of motion using Newton's $f = ma$ law.

Lagrangian dynamics: 2R open chain



- Consider a **2R open chain**, with links of masses m_1, m_2 respectively. To simplify things, we assume that the masses are concentrated at the ends of links.
- We take the **joint positions (θ_1, θ_2) for generalized coordinates**, and **(τ_1, τ_2) , the torques applied at the joints, as generalized forces**. Note that $\tau^\top \dot{\theta}$ is the power dissipated by the torques.
- We now need to derive the Lagrangian. To this end, we need the position and velocity of the masses.



$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} L_1 \cos \theta_1 \\ L_1 \sin \theta_1 \end{bmatrix} \text{ and } \begin{bmatrix} \dot{x}_1 \\ \dot{y}_1 \end{bmatrix} = \begin{bmatrix} -L_1 \sin \theta_1 \\ L_1 \cos \theta_1 \end{bmatrix} \dot{\theta}_1.$$

For link 2:

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2) \\ L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2) \end{bmatrix} \text{ and } \begin{bmatrix} \dot{x}_2 \\ \dot{y}_2 \end{bmatrix} = \begin{bmatrix} -L_1 \sin \theta_1 - L_2 \sin(\theta_1 + \theta_2) & -L_2 \sin(\theta_1 + \theta_2) \\ L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2) & L_2 \cos(\theta_1 + \theta_2) \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}.$$

- Using the relations of the previous slide, we obtain the **kinetic energy** of the links:

$$K_1 = \frac{1}{2}m_1(\dot{x}_1^2 + \dot{y}_1^2) = \frac{1}{2}m_1L_1^2\dot{\theta}_1^2$$

$$\begin{aligned}K_2 &= \frac{1}{2}m_2(\dot{x}_2^2 + \dot{y}_2^2) \\&= \frac{1}{2}m_2 \left((L_2^2 + 2L_1L_2 \cos \theta_2 + L_1^2)\dot{\theta}_1^2 + 2(L_2^2 + L_1L_2 \cos \theta_2)\dot{\theta}_1\dot{\theta}_2 \right. \\&\quad \left. + L_2^2\dot{\theta}_2^2 \right)\end{aligned}$$

- The **potential energies** of the links are

$$P_1 = m_1gy_1 = mgL_1 \sin \theta_1$$

$$P_2 = m_2gy_2 = m_2g(L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2))$$

- The **equations of motion** are $\tau_i = \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_i} - \frac{\partial L}{\partial \theta_i}$, $i = 1, 2$. This yields here

$$\begin{aligned}\tau_1 &= (\mathfrak{m}_1 L_1^2 + \mathfrak{m}_2 (L_1^2 + 2L_1 L_2 \cos \theta_2 + L_2^2)) \ddot{\theta}_1 \\ &\quad + \mathfrak{m}_2 (L_1 L_2 \cos \theta_2 + L_2^2) \ddot{\theta}_2 - \mathfrak{m}_2 L_1 L_2 \sin \theta_2 (2\dot{\theta}_1 \dot{\theta}_2 + \dot{\theta}_2^2) \\ &\quad + (\mathfrak{m}_1 + \mathfrak{m}_2) L_1 g \cos \theta_1 + \mathfrak{m}_2 g L_2 \cos(\theta_1 + \theta_2), \\ \tau_2 &= \mathfrak{m}_2 (L_1 L_2 \cos \theta_2 + L_2^2) \ddot{\theta}_1 + \mathfrak{m}_2 L_2^2 \ddot{\theta}_2 + \mathfrak{m}_2 L_1 L_2 \dot{\theta}_1^2 \sin \theta_2 \\ &\quad + \mathfrak{m}_2 g L_2 \cos(\theta_1 + \theta_2).\end{aligned}$$

- We can write the above equations as

$$\tau = M(\theta) \ddot{\theta} + \underbrace{c(\theta, \dot{\theta}) + g(\theta)}_{h(\theta, \dot{\theta})}$$

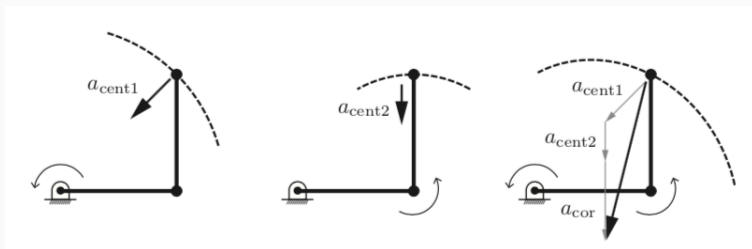
with the definitions

$$\begin{aligned}M(\theta) &= \begin{bmatrix} \mathfrak{m}_1 L_1^2 + \mathfrak{m}_2 (L_1^2 + 2L_1 L_2 \cos \theta_2 + L_2^2) & \mathfrak{m}_2 (L_1 L_2 \cos \theta_2 + L_2^2) \\ \mathfrak{m}_2 (L_1 L_2 \cos \theta_2 + L_2^2) & \mathfrak{m}_2 L_2^2 \end{bmatrix} \\ c(\theta, \dot{\theta}) &= \begin{bmatrix} -\mathfrak{m}_2 L_1 L_2 \sin \theta_2 (2\dot{\theta}_1 \dot{\theta}_2 + \dot{\theta}_2^2) \\ \mathfrak{m}_2 L_1 L_2 \dot{\theta}_1^2 \sin \theta_2 \end{bmatrix}, \\ g(\theta) &= \begin{bmatrix} (\mathfrak{m}_1 + \mathfrak{m}_2) L_1 g \cos \theta_1 + \mathfrak{m}_2 g L_2 \cos(\theta_1 + \theta_2) \\ \mathfrak{m}_2 g L_2 \cos(\theta_1 + \theta_2) \end{bmatrix},\end{aligned}$$

- The matrix $M(\theta)$ is **symmetric and positive definite**. It is called the **mass matrix**.
- The vector $c(\theta, \dot{\theta})$ contains the **centripetal and Coriolis forces/torques**, and $g(\theta)$ contains the **gravitational forces/torques**.
- We could have obtained the same equations again from $f = ma$

$$f_1 = \begin{bmatrix} f_{x1} \\ f_{y1} \\ f_{z1} \end{bmatrix} = m_1 \begin{bmatrix} \ddot{x}_1 \\ \ddot{y}_1 \\ \ddot{z}_1 \end{bmatrix} = m_1 \begin{bmatrix} -L_1\dot{\theta}_1^2 c_1 - L_1\ddot{\theta}_1 s_1 \\ -L_1\dot{\theta}_1^2 s_1 + L_1\ddot{\theta}_1 c_1 \\ 0 \end{bmatrix},$$
$$f_2 = m_2 \begin{bmatrix} -L_1\dot{\theta}_1^2 c_1 - L_2(\dot{\theta}_1 + \dot{\theta}_2)^2 c_{12} - L_1\ddot{\theta}_1 s_1 - L_2(\ddot{\theta}_1 + \ddot{\theta}_2) s_{12} \\ -L_1\dot{\theta}_1^2 s_1 - L_2(\dot{\theta}_1 + \dot{\theta}_2)^2 s_{12} + L_1\ddot{\theta}_1 c_1 + L_2(\ddot{\theta}_1 + \ddot{\theta}_2) c_{12} \\ 0 \end{bmatrix}$$

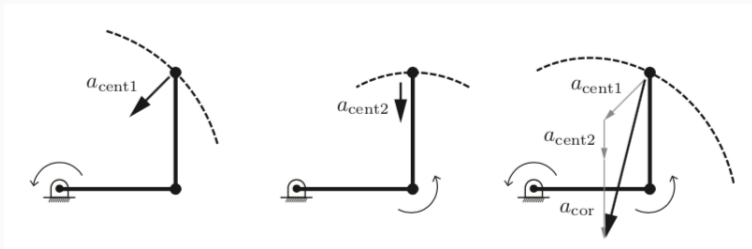
- Note that since $(\hat{x}, \hat{y})$ is an **inertial frame**, we have equations $\ddot{x} = \dots, \ddot{y} = \dots$. The frame $(\hat{\theta}_1, \hat{\theta}_2)$ is not inertial, hence $\ddot{\theta} = 0$ does not mean there is zero acceleration (i.e., $\ddot{x}, \ddot{y}$ are not necessarily zero when $\ddot{\theta}$ is.).



- A zero acceleration in a non-inertial frame does not imply a zero acceleration in an inertial frame.
- Consider the arm in position $(\theta_1, \theta_2) = (0, \pi/2)$. Assuming $\ddot{\theta} = 0$, we have

$$\begin{bmatrix} \ddot{x}_2 \\ \ddot{y}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} -L_1 \dot{\theta}_1^2 \\ -L_2 \theta_1^2 - L_2 \dot{\theta}_2^2 \end{bmatrix}}_{\text{centripetal}} + \underbrace{\begin{bmatrix} 0 \\ -2L_2 \dot{\theta}_1 \dot{\theta}_2 \end{bmatrix}}_{\text{Coriolis}}$$

- Quadratic terms $\dot{\theta}_i^2$ are called **centripetal** terms, the mixed quadratic terms $\dot{\theta}_1 \dot{\theta}_2$ the **Coriolis** terms.



- If $\dot{\theta}_2 = 0$, no Coriolis and centrifugal accel. is $(-L_1\dot{\theta}_1^2, -L_2\dot{\theta}_1^2)$. Similarly, $\dot{\theta}_1 = 0$, no Coriolis and centrifugal accel. is $(0, -L_2\dot{\theta}_2^2)$. These accelerations keep the mass rotating around the joints 1 and 2 respectively.
- The Coriolis force appears if both $\dot{\theta}_i$ are non-zero. Note that its sign depends on the signs of the $\dot{\theta}_i$'s.

- For a general open chain with n links, we take the link angles θ_i as generalized coordinates and the corresponding torques τ_i as generalized forces.
- The kinetic energy can always be written as

$$K(\theta, \dot{\theta}) = \dot{\theta}^\top M(\theta) \dot{\theta}$$

for an appropriately defined mass matrix $M(\theta)$.

- The dynamics equation are then

$$\tau_i = \frac{d}{dt} \frac{\partial L}{\partial \dot{\theta}_i} - \frac{\partial L}{\partial \theta_i}$$

for $L = K - P$ with $P(\theta)$ the potential energy of the system.

- Written explicitly, we have

$$\tau_i = \sum_{j=1}^n m_{ij}(\theta) \ddot{\theta}_j + \sum_{j,k=1}^n \Gamma_{ijk}(\theta) \dot{\theta}_j \dot{\theta}_k + \frac{\partial P}{\partial \theta_i}$$

where

$$\Gamma_{ijk}(\theta) = \frac{1}{2} \left(\frac{\partial m_{ij}}{\partial \theta_k} + \frac{\partial m_{ik}}{\partial \theta_j} - \frac{\partial m_{jk}}{\partial \theta_i} \right)$$

are the Christoffel's symbols of the first kind.

- This dynamics is also written as

$$\tau = M(\theta) \ddot{\theta} + C(\theta, \dot{\theta}) \dot{\theta} + g(\theta),$$

where M is the mass matrix, and $C(\theta, \dot{\theta}) \in \mathbb{R}^{n \times n}$ is the matrix with entries

$$c_{ij} = \sum_{k=1}^n \Gamma_{ijk}(\theta) \dot{\theta}_k.$$

It is called the Coriolis matrix.